Haifuraiya futureGEO Proposal

Open Research Institute

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Who, What, When, Where, Why

- Who is ORI?
- Open Research Institute is a nonprofit research and development firm incorporated in California, USA, with international participation.
- All volunteer
- Volunteers, Project Leads, Board of Directors
- Community of 2000+, ~12 projects, 250 active on Slack, ~5-10 developing at any one time.
- https://openresearch.institute

Who, What, When, Where, Why

- What do we do?
- Open Source Digital Radio Research and Development
- All work published at https://github.com/OpenResearchInstitute
- Technical and Regulatory

Who, What, When, Where, Why

- When?
- Founded in 2019
- Done on all volunteer time, with some volunteers active daily and many active weekly or monthly.

Who, What, When, Where, Why

- Where?
- Volunteers come from all over the world (most are US-based, Africa is the most underrepresented)
- We demonstrate regularly at conferences and events.
- “It doesn’t work until it works over the air”
- Next major live demonstration is DEFCON in Las Vegas, NV, USA in August 2026. ORI will have an Open Source Digital Radio exhibit at RF Village and give a talk about Earth-Venus-Earth communications attempt in October 2026.

Who, What, When, Where, Why

- Why?
- We focus on work that 1) is useful and 2) doesn't *need* to be proprietary
- For workforce development
- For educational development
- For professional development
- For personal development
- To support organizations with Open Source technical or regulatory solutions (such as AMSAT-DL)

What is Haifuraiya? ハイフライヤ

FDMA digital Opulent Voice uplinks, on-board processing, TDM digital DVB-S2 downlink

- A digitally regenerative, software-defined GEO amateur radio payload is the right next step after QO-100.
- Opulent Voice protocol uplink provides truly excellent voice quality. Far superior to any amateur digital voice today. Baseline is 16 kbps OPUS codec.
- DVB-S2 downlink provides proven, powerful, and flexible downlink.
- No second clunky packet radio mode for text or general data with Opulent Voice. Data and voice are nicely integrated.
- Network stack is in the protocol (RTP, UDP, IP, TCP).
- Not a leap of faith. This is working and actively developed open-source hardware and software.

Working Reference Designs

Major or Top-Level Repository Listings - more underneath

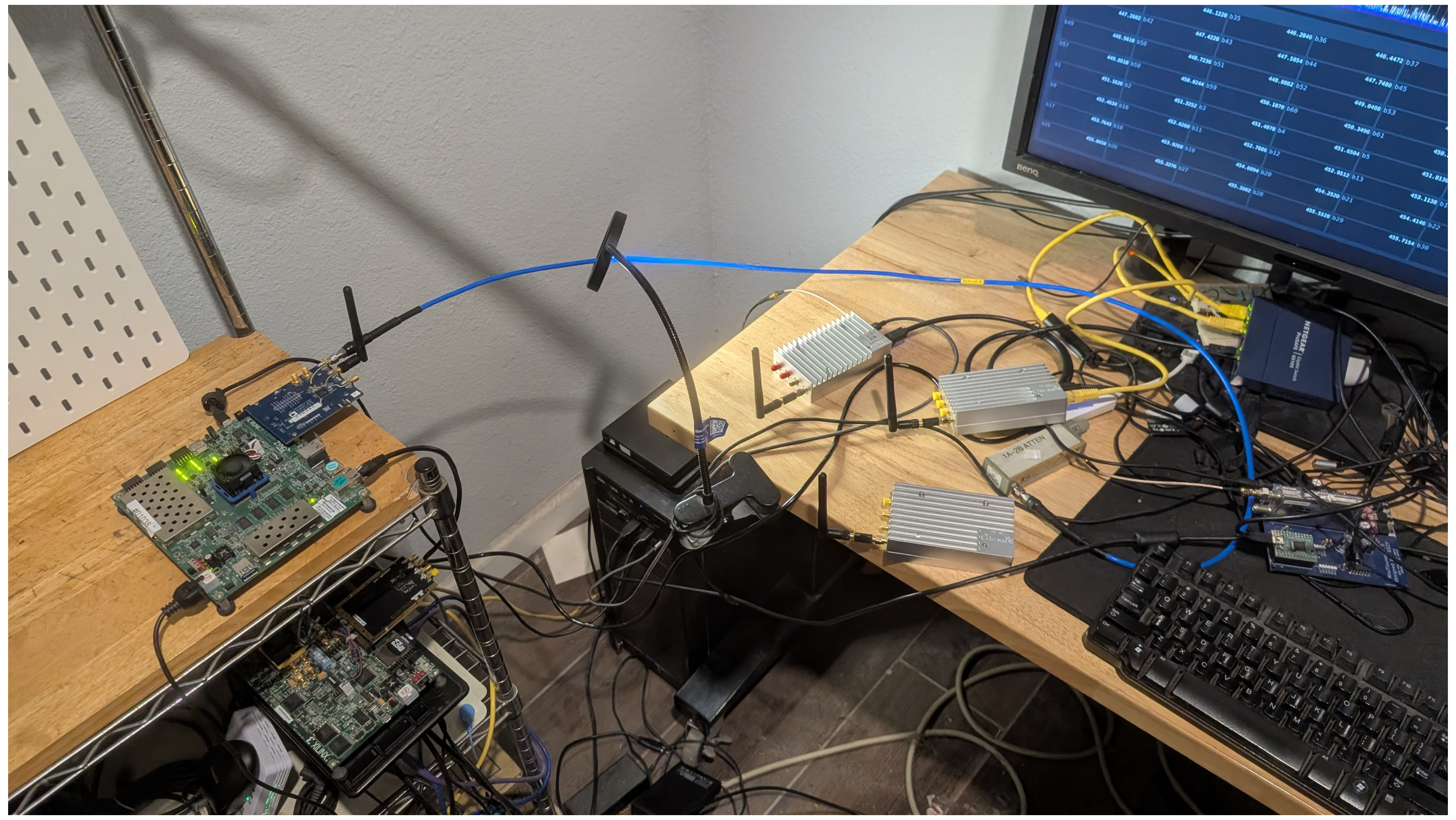
- https://github.com/OpenResearchInstitute/pluto_msk
- https://github.com/OpenResearchInstitute/dialogus
- https://github.com/OpenResearchInstitute/interlocutor
- https://github.com/OpenResearchInstitute/locus
- https://github.com/OpenResearchInstitute/opv-cxx-demod
- https://github.com/OpenResearchInstitute/dvb_fpga
- https://github.com/OpenResearchInstitute/Mode-Dynamic-Transponder

Reference Designs

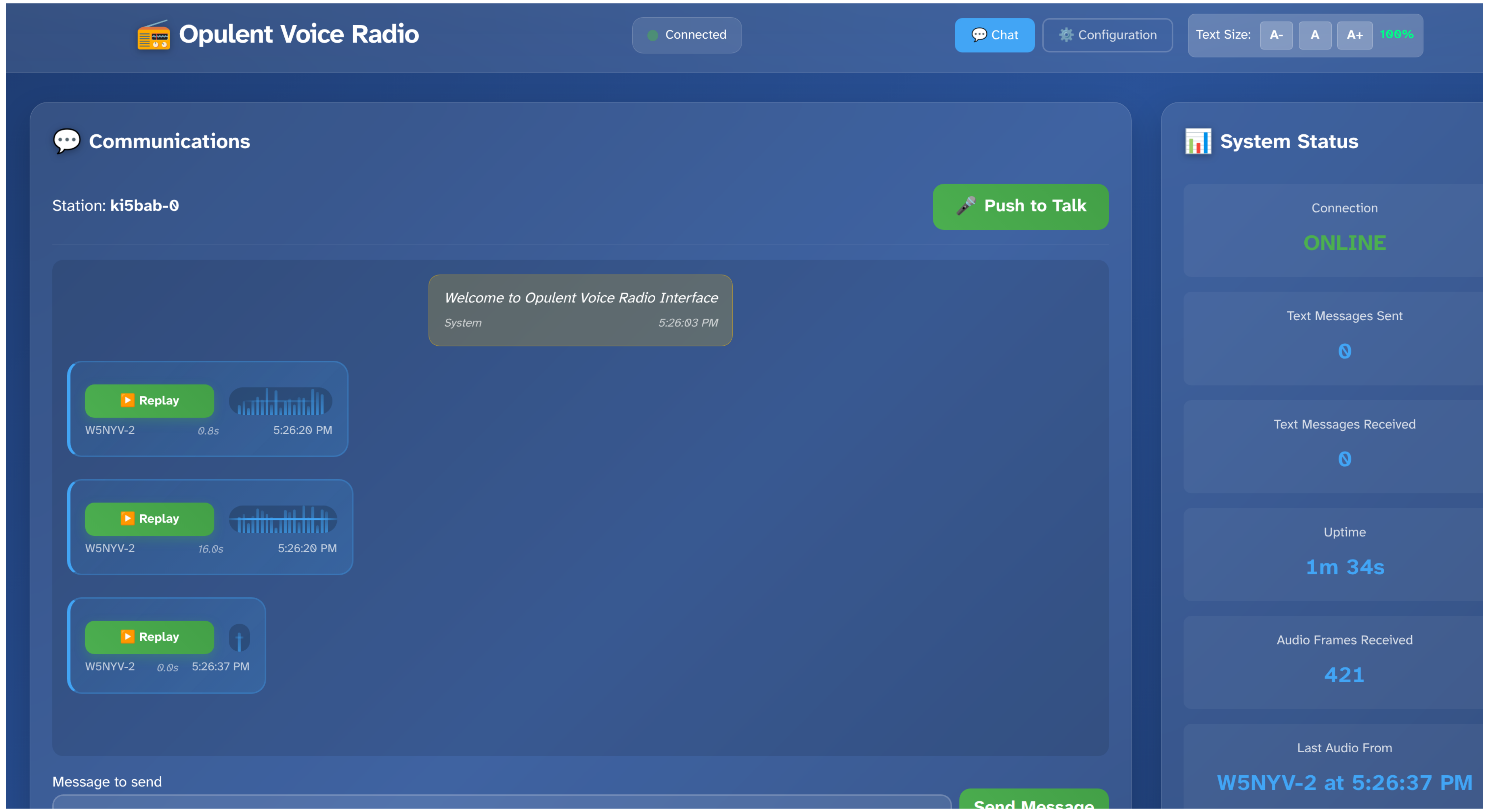
What does it look like when it's all put together?

https://youtu.be/UABDaQ6oRlw

Here is one of the three current implementations, demonstrated live in a YouTube video.



Interlocutor lets you play back received transmissions, optional text to speech, and optional audio transcriptions.



Accessibility built in from the start.

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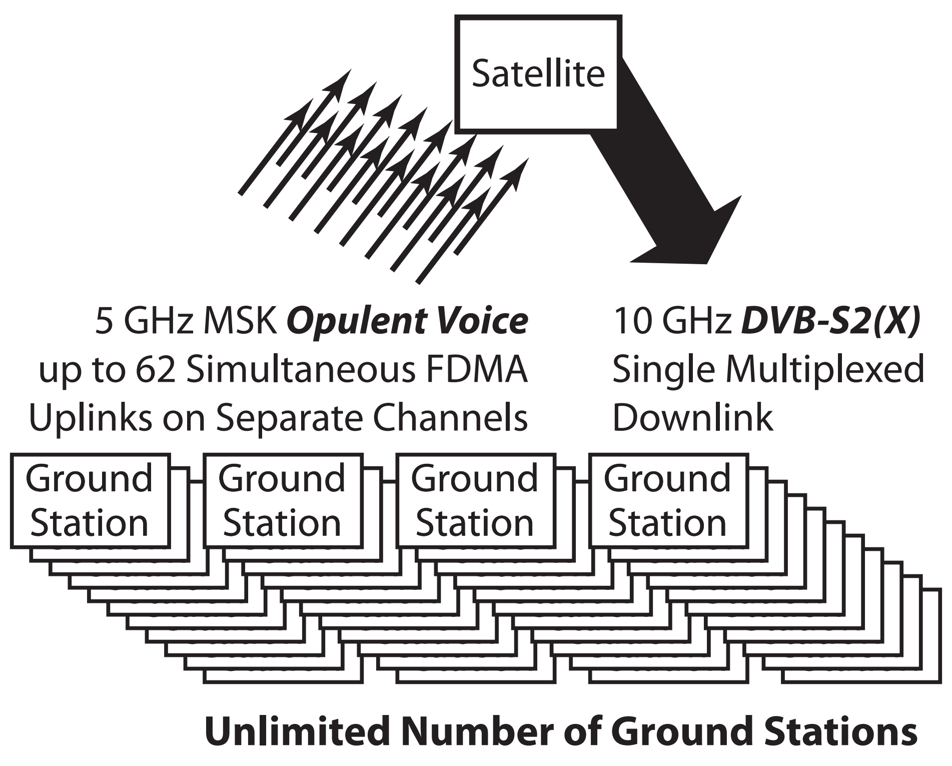
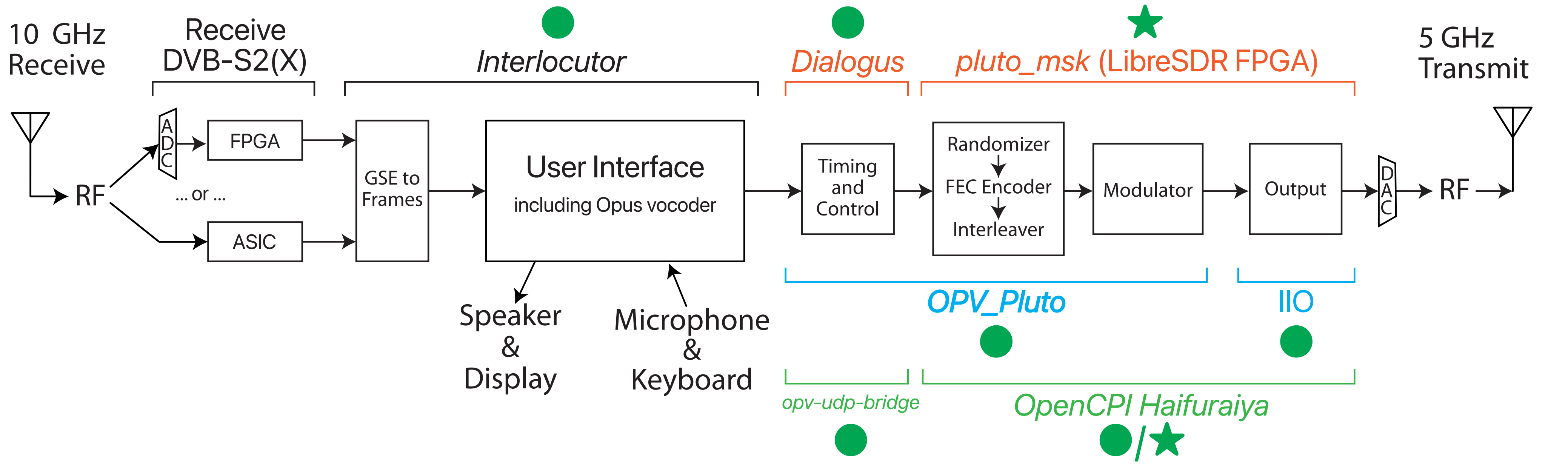
ORI *Haifuraiya* Digital Satellite System

Haifuraiya Vocabulary and links

System Overview



Three Ground Station Implementations (so far)



★ = VHDL FPGA module is implemented and largely working.

● = Software module is implemented and largely working.

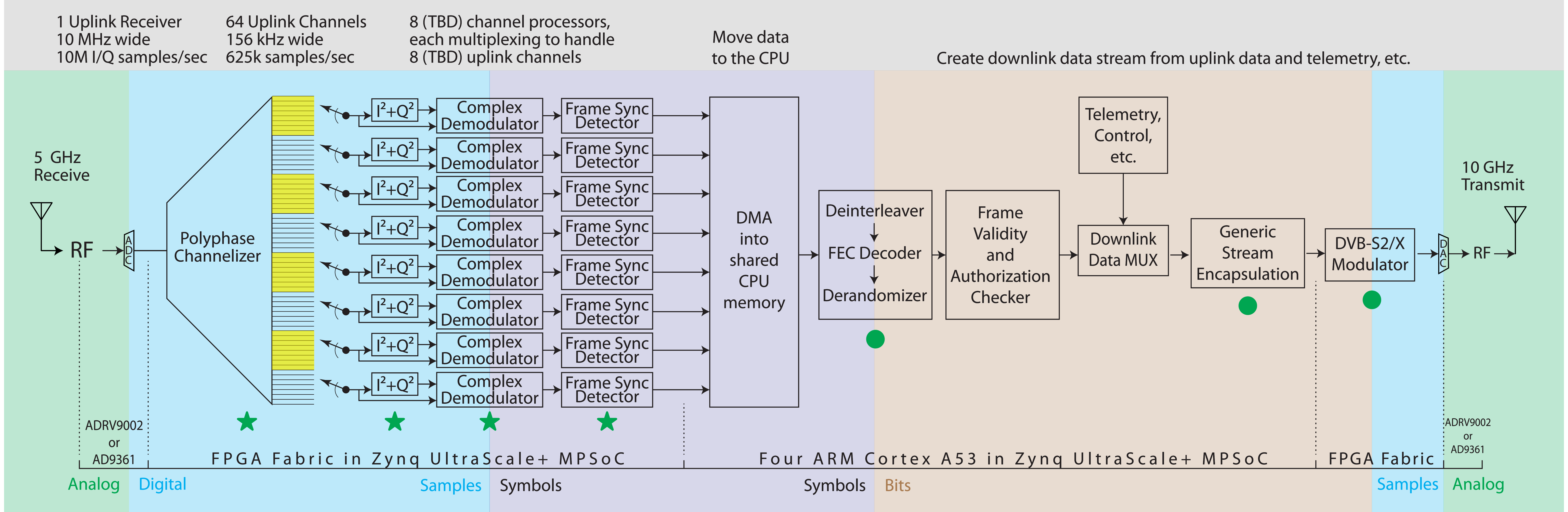
Open Research Institute (ORI) is a non-profit research and development organization which provides all of its work to the general public under the principles of Open Source and Open Access to Research.

100% Open Source: Everything in these designs is released under recognized open source licenses. Hardware is compliant with the Open Source Hardware Definition and licensed under the CERN Open Hardware License Version 2 Permissive. Software is licensed under the GNU General Public License 3.0, or compatible. Both are published on public **git** repositories, early and often.

Satellite Transponder Architectures

One of two separate but similar implementations shown.

Everything in the Haifuraiya transponder between the receive low noise amplifier and the transmit filters and power amplifer runs on a subset of the Xilinx ZCU102 MPSoC development board (mainly the Zynq UltraScale+ device and its direct support circuits) plus the Analog Devices ADRV9002 or AD9361 RF transceiver eval board.



Questions?